

Simulating Biomarker-Treatment Interactions: Correlation-Based vs. Mean-Structure Approaches

pmsimstats2025 Project Notes

1. The Problem

The pmsimstats2025 simulation and the original Hendrickson et al. (2020) codebase use fundamentally different mechanisms to generate the biomarker-treatment interaction – the statistical signal that the simulation is designed to detect. This document reviews the two approaches, surveys the relevant literature, and evaluates the pros and cons of each for the N-of-1 aggregated trial context.

The biomarker-treatment interaction is a predictive (not prognostic) effect: participants with higher biomarker values respond *differently* to treatment than those with lower values. The analysis model tests this via an interaction term (`Dbc:bm_centered` or `treatment:bm_centered`). The question is how the data-generating process (DGP) should encode this interaction so that the simulation faithfully represents the clinical scenario.

2. Approach 1: Mean-Structure (Deterministic Modulation)

2.1 Mechanism

The biomarker enters the DGP as a fixed coefficient in the linear predictor. A participant's biological response to drug is a deterministic function of their biomarker value:

```
Y_i = baseline + BR_i + ER_i + TR_i + noise  
BR_i = f(drug_exposure) * (1 + beta_TB * bm_centered_i)
```

A participant 1 SD above the biomarker mean receives exactly $(1 + \text{beta_TB})$ times the treatment effect of the average participant. The residual noise adds variability but does not modulate the interaction. The biomarker-treatment relationship is known with certainty conditional on the biomarker value.

2.2 pmsimstats2025 implementation

In `carryover_power_simulation.R` (lines 345–347):

```
effective_BR_rate <- BR_rate * (1 + bm_mod * bm_centered)  
br_on_drug <- weeks_on_drug * effective_BR_rate
```

The sigma matrix applies a constant `c.bm` to all BR timepoints regardless of treatment status (lines 148–153), plus half-strength correlations to ER and TR that the original does not have. The biomarker-BR correlation in the covariance structure serves only to generate realistic between-participant variability; it does not encode the treatment-dependent interaction.

2.3 Literature using this approach

Nearly all published biomarker-stratified trial simulations use the mean-structure approach:

Riecke et al. (2018). Simulated Cox regression with $\text{beta_TxB} * T * B$ in the log-hazard. Three interaction strengths ($\text{beta_TxB} = 0, 0.095, 0.285$). Found that including prognostic covariates increases power to detect the interaction and reduces bias. 500 subjects, 1000 simulations, 108 scenarios. (DOI: 10.1186/s12874-018-0457-9)

Haller et al. (2019). Simulated survival outcomes with deterministic hazard-rate functions of biomarker Z and treatment T. Six interaction shapes (null, constant HR, linear, nonlinear monotonic, tail effect, U-shaped). Compared seven analysis methods (median split, quartile split, optimal cutpoint, STEPP, Cox linear, MFPI, LPLB). Cox regression with linear interaction was most efficient for monotonic shapes; MFPI and LPLB outperformed for complex nonlinear interactions. (DOI: 10.1159/000500436)

Langbaum et al. (2014). Alzheimer’s disease prevention trial enrichment. Treatment effect applied only to biomarker-positive (at-risk) participants; non-at-risk participants received zero treatment effect. Found that ignoring sample heterogeneity (different variance components for at-risk vs. not-at-risk groups) overestimates power. (DOI: 10.1016/j.jalz.2013.12.019)

Friede, Parsons, and Stallard (2012). Adaptive subgroup selection designs. Treatment effects specified as fixed mean differences for different biomarker-defined subgroups. Simulation operates at the level of sufficient statistics (multivariate normal test statistics), not individual patient data. (DOI: 10.1002/sim.5541)

Blackston et al. (2019). Aggregated N-of-1 vs. RCT comparison. Treatment effect is fixed and homogeneous across all participants; no biomarker-treatment interaction is modeled. Carryover is modeled as fixed increments. (DOI: 10.3390/healthcare7040137)

2.4 Pros

1. **Direct control over effect size.** The parameter `beta_TB` maps directly to the interaction coefficient that the analysis model estimates. Power calculations are analytically tractable because the true parameter is known exactly. For a given `beta_TB`, `Var(B)`, and residual variance, the expected power can be derived in closed form.
2. **Transparent DGP-analysis alignment.** The DGP and the analysis model share the same parametric form. If the analysis model is $Y \sim T * B$, the simulation generates data from the same family. There is no model misspecification by construction, making it straightforward to verify that the simulation is calibrated correctly (i.e., that `mean(beta_hat) = beta_TB` over replications).
3. **Standard in the literature.** Nearly every simulation paper outside Hendrickson uses this approach, making results directly comparable to published power estimates for biomarker-stratified designs.
4. **Computationally simple.** No path-specific covariance matrices are needed. One set of parameters generates all participants regardless of treatment assignment. The sigma matrix can be pre-built and cached once per design.

2.5 Cons

1. **Overly optimistic about biomarker predictive utility.** Given a biomarker value, the treatment effect is known with certainty (up to residual noise). In clinical practice, biomarkers are imperfect predictors: two patients with identical biomarker values can have very different treatment responses for reasons unrelated to the biomarker (genetic background, comorbidities, adherence). The mean-structure approach cannot represent this heterogeneity.
2. **No high-biomarker non-responders (except through noise).** A patient 2 SD above the biomarker mean always has a large treatment effect mean. The only way they can be a non-responder is if the residual error overwhelms the signal. Clinically, non-response despite favorable biomarker status is common and may reflect biological complexity that the mean-structure approach does not capture.
3. **DGP-analysis alignment is both a feature and a limitation.** Because the simulation generates data from the same model family that the analysis uses, it cannot reveal the consequences of model misspecification. If the real-world interaction operates through a different mechanism (e.g., covariance-mediated), the simulation will overestimate the power of the standard analysis model.
4. **Treatment-dependent variance is absent.** In the mean-structure approach, the residual variance is the same whether a participant is on drug or off drug, and whether the biomarker is high or low. In clinical data, treatment response variability often depends on treatment status and patient characteristics.

3. Approach 2: Correlation-Based (Covariance Structure)

3.1 Mechanism

The biomarker and the biological response component are drawn jointly from a multivariate normal distribution. The interaction is encoded as a correlation between the biomarker and the treatment-response component, with the correlation depending on treatment status:

```
(B_i, BR_i(t)) ~ MVN(mu, Sigma)
```

where:

```
corr(B, BR(t)) = c.bm      when on drug (tod > 0)
corr(B, BR(t)) = scaled    when off drug with carryover
corr(B, BR(t)) = 0         when off drug, no carryover
```

The biomarker does not appear in the mean of BR. Instead, participants with higher biomarker draws *tend* to get higher BR draws through the joint distribution. The tendency is controlled by `c.bm`, and the prediction is inherently imperfect: even at `c.bm = 0.6`, approximately 64% of the variance in BR is unexplained by the biomarker.

3.2 Hendrickson’s original implementation

In `generateData.R` (lines 84–163), the biological response means are computed via a modified Gompertz function of cumulative time on drug (`tod`):

```
brmeans <- modgompertz(d$tod, rp$max, rp$disp, rp$rate)
```

The biomarker-BR correlation is then set in the correlation matrix, with the Ron Thomas modification (lines 146–162):

```
for (p in 1:nP) {
  n1 <- paste(trialdesign$timeptname[p], "br", sep = ".")
  if (p > 1) {
    n0 <- paste(trialdesign$timeptname[p - 1], "br", sep = ".")
    mm1 <- means[which(n1 == labels)]
    mm0 <- means[which(n0 == labels)]
    correlations["bm", n1] <- ifelse(
      brtest[p],
      ifelse(brmeans[p] == 0, 0, (mm1 / mm0) * c.bm),
      c.bm
    )
  }
}
```

The logic:

- `brtest[p]` is TRUE when the raw BR mean (from Gompertz of `tod`) is zero – i.e., when the participant has zero cumulative drug exposure at that timepoint.
- **On-drug timepoints** (`brtest` FALSE, `tod > 0`): `corr(bm, BR) = c.bm`. Full correlation. High-biomarker participants tend to draw higher BR values.
- **Off-drug timepoints, no carryover** (`brtest` TRUE, `brmeans = 0`): `corr(bm, BR) = 0`. No correlation. The biomarker has no predictive value when there is no drug effect to predict.
- **Off-drug timepoints, with carryover** (`brtest` TRUE, `brmeans > 0` after carryover adjustment): `corr(bm, BR) = (mm1/mm0) * c.bm`. Scaled correlation proportional to the ratio of current to previous BR mean. As carryover decays, the correlation approaches zero.

Crucially, the first timepoint (`p = 1`) is skipped entirely (the `p > 1` guard), so the first timepoint’s BR has zero correlation with the biomarker regardless of treatment status.

Because the correlation structure depends on the treatment sequence, each path through the trial design produces a different covariance matrix. The original generates data separately for each path using path-specific sigma matrices, then merges the paths into a single dataset for analysis.

3.3 Literature using this approach

Hendrickson et al. (2020) is the only simulation study identified in the biomarker-treatment interaction literature that uses this approach. The paper describes the simulation framework but does not explicitly discuss the choice between correlation-based and mean-structure interaction encoding.

3.4 Pros

1. **Clinical realism.** Biomarkers are imperfect predictors. The correlation-based approach naturally generates high-biomarker non-responders and low-biomarker responders at rates controlled by `c.bm`. At `c.bm = 0.6`, the biomarker explains 36% of BR variance; the remaining 64% represents unmeasured sources of treatment response heterogeneity. This is consistent with the empirical observation that even the best clinical biomarkers explain only a modest fraction of treatment response variability.
2. **Treatment-dependent covariance is biologically principled.** The biomarker should predict response only when the drug is active. When off drug ($BR = 0$), there is nothing for the biomarker to predict, so $\text{corr}(bm, BR) = 0$. The correlation-based approach encodes this directly in the covariance structure: the statistical relationship between biomarker and outcome changes with treatment status. The mean-structure approach achieves this indirectly through the $T * B$ interaction term, which is algebraically equivalent for the conditional mean but does not produce the same conditional variance structure.
3. **Tests the analysis model under realistic misspecification.** Because the DGP and the analysis model have different parametric forms (MVN covariance vs. LME interaction coefficient), the simulation reveals how well the standard analysis model recovers the true signal when the data-generating mechanism is more complex than the analysis assumes. This is the typical situation in real trials, where the true DGP is never exactly the model being fitted.
4. **Consistent with Senn’s variance decomposition.** Senn (2016) decomposes treatment response heterogeneity into patient-by-treatment interaction variance (ψ^2). A biomarker in the correlation-based DGP explains a fraction of ψ^2 proportional to $c.bm^2$, leaving residual heterogeneity. In the mean-structure approach, ψ^2 is entirely determined by $\text{beta_TB}^2 * \text{Var}(B)$, with no residual treatment heterogeneity beyond noise. The correlation-based approach is therefore the more conservative (and arguably more honest) representation of biomarker utility.

3.5 Cons

1. **Indirect control over the detectable effect size.** The analysis model estimates `beta_TB` (the interaction coefficient), but the DGP does not contain `beta_TB` as a parameter. The relationship between `c.bm` and the expected `beta_hat_TB` depends on the variances of `B` and `BR`, the Gompertz curve shape, the number of on-drug timepoints, the autocorrelation structure, and the correlation between factors – a complex mapping that makes analytic power calculation difficult. In practice, the expected `beta_hat_TB` must be determined empirically by running the simulation and averaging the estimates.
2. **Path-specific covariance matrices required.** Because the `bm-BR` correlation depends on treatment status, each path through the trial design produces a different covariance matrix. For the Hybrid design with 4 paths, this requires 4 sigma matrices per parameter combination. Pre-computation and caching become more complex, particularly because the `BR` means (which determine the Ron Thomas scaling) depend on the response parameters (Gompertz max, displacement, rate), not just the design geometry.
3. **Harder to validate.** With the mean-structure approach, one can verify that $\text{mean}(\text{beta_hat}) = \text{beta_TB}$ over replications to confirm calibration. With the correlation-based approach, the expected `beta_hat`

is a derived quantity that depends on many parameters jointly. Validation requires either analytic derivation of the expected interaction coefficient (nontrivial) or large-sample simulation calibration.

4. **Unique to Hendrickson.** No other published simulation study in the biomarker-stratified trial literature uses this approach, limiting comparability with published power estimates. Results from the correlation-based DGP cannot be directly compared to power estimates from Riecke et al., Haller et al., or other standard references.
5. **The Ron Thomas scaling is fragile.** The `mm1/mm0` ratio depends on the Gompertz means at consecutive timepoints. When the denominator (`mm0`) is small (early timepoints with short drug exposure), the ratio can be large, producing correlation values greater than `c.bm`. When both means are near zero (off-drug, minimal carryover), the ratio is numerically unstable. The `brtest` guard handles the exact-zero case but does not address near-zero instabilities.

4. The Conceptual Bridge

Senn’s (2016) variance decomposition framework clarifies that the two approaches are not alternatives but rather represent different assumptions about the same variance partition.

Consider the treatment effect for patient i :

$$\tau_i = \bar{\tau} + \gamma B_i + u_i$$

where $\bar{\tau}$ is the average treatment effect, γ is the biomarker moderation coefficient, B_i is the biomarker value, and u_i captures residual treatment heterogeneity unexplained by the biomarker.

- **Mean-structure approach:** Sets $\text{Var}(u_i) = 0$. The biomarker explains all systematic treatment heterogeneity. Any remaining variation in treatment response is pure noise (the residual ε in the outcome model). The total patient-by-treatment interaction variance is $\psi^2 = \gamma^2 \text{Var}(B)$.
- **Correlation-based approach:** Allows $\text{Var}(u_i) > 0$. The biomarker explains only a fraction of treatment heterogeneity, controlled by `c.bm`. The total patient-by-treatment interaction variance is $\psi^2 = \gamma^2 \text{Var}(B) + \text{Var}(u)$, where γ is implied by the covariance structure rather than specified directly.

At `c.bm = 1.0`, the correlation-based approach converges to the mean-structure approach: the biomarker perfectly predicts BR, and the stochastic correlation collapses to a deterministic relationship. At `c.bm < 1.0`, the correlation-based approach generates more residual heterogeneity, which attenuates the detectable interaction and reduces power relative to the mean-structure approach for the same nominal biomarker-response association.

This attenuation is not a defect but a realistic representation: if the biomarker explains only 36% of treatment response variance (`c.bm = 0.6`), then 64% of the variance appears as noise to the analysis model, and the power to detect the interaction is correspondingly lower.

5. Implications for Carryover Modeling

The choice of interaction mechanism affects how carryover contaminates the analysis in distinct ways.

5.1 Under the mean-structure approach

At off-drug timepoints with carryover, the DGP generates:

$$\text{BR_off} = \text{br_at_discont} * \text{Dbc} * (1 + \text{bm_mod} * \text{bm_centered})$$

The biomarker modulation is present at full strength whenever `Dbc > 0`. The naive model (`treatment = 0` at these timepoints) attributes this biomarker-modulated residual drug response to noise. The interaction estimate is attenuated because the on/off contrast is contaminated.

5.2 Under the correlation-based approach

At off-drug timepoints with carryover, the DGP generates BR draws from an MVN where $\text{corr}(\text{bm}, \text{BR})$ is scaled by the Ron Thomas ratio ($\text{mm1}/\text{mm0}$). As carryover decays, the scaled correlation approaches zero. The biomarker’s predictive value at off-drug timepoints decays in parallel with the drug effect. The naive model’s misspecification is proportional to the *covariance* between the biomarker and the off-drug BR, which is a function of both the carryover level and the correlation scaling – a more complex interaction than the mean-structure approach produces.

5.3 The practical consequence

For the two-scenario comparison (WITH vs. WITHOUT carryover modeling), the power gap between the two analysis models will be *smaller* under the correlation-based DGP than under the mean-structure DGP, because the correlation-based approach generates weaker interaction signal at off-drug timepoints (the correlation is scaled down, not just the mean). This means the pmsimstats2025 simulation, which uses the mean-structure approach, may *overestimate* the benefit of modeling carryover relative to what the original Hendrickson DGP would show.

6. Summary Comparison

Feature	Mean-Structure	Correlation-Based
Interaction location	Fixed coefficient in the linear predictor	Joint covariance of biomarker and BR
Nature of prediction	Deterministic given biomarker	Stochastic; probabilistic tendency
High-BM non-responders	Only through residual noise	Natural; frequency controlled by c.bm
Effect size control	Direct (beta_TB)	Indirect (c.bm, variances, Gompertz)
Analytic power formulas	Standard; tractable	Difficult; requires simulation
Sigma matrices needed	1 per design (path-invariant)	1 per path per design
DGP-analysis alignment	Exact (same parametric family)	Misspecified (realistic)
Treatment-dependent variance	No	Yes (correlation varies with drug status)
Literature precedent	Nearly universal	Hendrickson (2020) only
Computational cost	Lower	Higher (path-specific sigma)
Biomarker utility estimate	Optimistic	Conservative

7. Recommendation for pmsimstats2025

The choice depends on the simulation’s purpose:

For faithful replication of Hendrickson (2020): The correlation-based approach is required. The original’s power estimates reflect the correlation-based mechanism with path-specific sigma matrices and the Ron Thomas scaling logic. Switching to mean-structure will produce systematically higher power for the same nominal c.bm, making comparison with the published heatmaps impossible. This is Phase 1 (Changes 1–2) in the comprehensive status document.

For the two-scenario comparison paper: Both approaches should be reported. The correlation-based DGP establishes that the two-scenario comparison result (modeling carryover recovers power) holds under Hendrickson’s original assumptions. The mean-structure DGP demonstrates generalizability: the same result holds under the more common simulation convention, facilitating comparison with the broader biomarker-stratified trial literature.

For methodological contribution: No published study compares the operating characteristics (power, type I error, bias, coverage) of these two simulation approaches for the same analysis model. Reporting both

approaches side by side, with the same analysis model and the same nominal biomarker-response association strength, would quantify how much the choice of DGP affects estimated power. This is a genuine gap in the simulation methodology literature.

8. References

- Hendrickson RC, Thomas RG, Schork NJ. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation. *Frontiers in Digital Health*. 2020;2:13. doi: 10.3389/fdgth.2020.00013
- Riecke BF, et al. A simulation study on estimating biomarker-treatment interaction effects in randomized trials with prognostic variables. *BMC Medical Research Methodology*. 2018;18:32. doi: 10.1186/s12874-018-0457-9
- Haller B, et al. A simulation study comparing different statistical approaches for the identification of predictive biomarkers. *Methods of Information in Medicine*. 2019;58(S01):e60-e79. doi: 10.1159/000500436
- Langbaum JB, et al. Simulating effects of biomarker enrichment on Alzheimer’s prevention trials. *Alzheimer’s & Dementia*. 2014;10(5 Suppl):S388-S395. doi: 10.1016/j.jalz.2013.12.019
- Friede T, Parsons N, Stallard N. A conditional error function approach for subgroup selection in adaptive clinical trials. *Statistics in Medicine*. 2012;31(30):4309-4320. doi: 10.1002/sim.5541
- Blackston JW, et al. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *Healthcare*. 2019;7(4):137. doi: 10.3390/healthcare7040137
- Senn S. Mastering variation: variance components and personalised medicine. *Statistics in Medicine*. 2016;35(7):966-977. doi: 10.1002/sim.6739
- Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-383. doi: 10.1177/0962280217726801
- Zhang Z, et al. The use of covariates and random effects in evaluating predictive biomarkers under a potential outcome framework. *Annals of Applied Statistics*. 2014;8(4):2336-2355. doi: 10.1214/14-AOAS773

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